San Francisco has one of the best public transportation systems in the United States. Here I am comparing the percentage of households in SF counties that own a vehicle to the percentage of different age groups and poverty rates in the area to see how these factors influence vehicle ownership.

These maps are classified based on percentage of households without a vehicle, percentage of seniors 65+, percentage of children age 0-17, and the poverty rate among SF counties. If you hover over a map, the GEOID will display allowing you to match it with the specific county, as well as the values associated with variable tract ID giving you the percentage of whatever attribute you are looking at. When you hover over one, the GEOID and values display across the other 3 maps as well, which makes it easy to compare data across each map at the same time.

It appears that across central/southern SF counties, there is a low percentage of households without a vehicle. Comparing this to our other maps, poverty rate is also lowest in central SF and percentage of children in highest across central/southern SF. From a family perspective, it makes sense that areas with a higher percentage of children under 18 years old have a higher percentage of vehicles per household as the children must be driven to school/extracurriculars/appointments and some may not be old enough to ride public transit/walk by themselves. Poverty rate is also low in this area which can be related to the fact that in order town a car, you need enough space to house it, and space is expensive in San Francisco since it is very limited. There is some overlap between high percentage of children and high poverty rate seen in southeast SF where there is also low vehicle ownership. To explain this, I believe there needs to be a further analysis done on access to public transportation and type of housing in the area. Seniors age 65+ are seen in high percentages scattered across central and up the western coast to northeast corner of SF. It is essentially the opposite of the poverty rate map, indicating that most seniors 65+ living in SF counties do not struggle with low-income, likely because they are retired in SF and have the wealth to remain or have lived in the area for enough time to establish a well-paying job that supports them. High percentages of Seniors 65+ can also be seen in areas with low percentages of households without a vehicle, indicating they Seniors 65+ likely own a vehicle and have a personal mode of transportation.

Moving forward, it would be interesting to analyze the public transportation in the area and compare households that are closer to public transportation with vehicle ownership and see how accessibility plays a role. Another way of furthering this study is to map residential housing with urban housing and see which area utilizes personal vehicles vs public transport.

Classifications schemes used include quantile, equal interval, and natural breaks. Quantile classification. Quantile classification creates classes that contain an equal number of features and is suited for linearly distributed data. This ensures that there are no empty classes or ones with few values compared to others. Problems associated with this are that final maps using Quantiles can be misleading, as similar features can be placed in different classes or features with widely different values can be placed into the same class. These issues can be minimized if the number of classes is increased. Equal interval classification scheme divides the range of attribute values into equal ranges allowing the specification of intervals and class breaks based on value range are automatically chosen. This classification works best for data ranges utilizing percentages such as the normalized datasets used for these maps. It also emphasizes the amount of the given attribute values compared to others to see examine the range of values. I find it visualizes the change in data across the 4 maps when you increase the number of classes. Natural breaks classifications scheme is based on natural groupings that represent the spread of data. The values are grouped in a way where similar values are grouped, and the differences between classes are maximized so visualizing the data is clear. Although natural breaks are good for analyzing differences in values, it is not useful for comparing multiple maps and should not be used for a valuable interpretation of the 4 maps provided.